Practice: Sampling Distributions and Standard Errors

Module 4 | Lesson 2 | Individuals versus statistics

Use the lesson for the checked case introductions and source limits. Submit a short Quarto report with calculations, units, assumptions, and interpretation.

1. For EAI, calculate the approximate salary SE at n = 30 and n = 120. What stays unchanged about individual salary SD?
2. Compute finite-population-corrected salary SEs at both sizes. Which sampling fraction exceeds 5%?
3. Calculate the approximate training-proportion SE at n = 30 and check the expected counts for a normal approximation.
4. Repeat the lesson's artificial mean simulation with n = 120. Label its model and compare empirical SD with theoretical SE.
5. Explain why the CLT does not justify a normal salary histogram or fix dependence between coworkers.